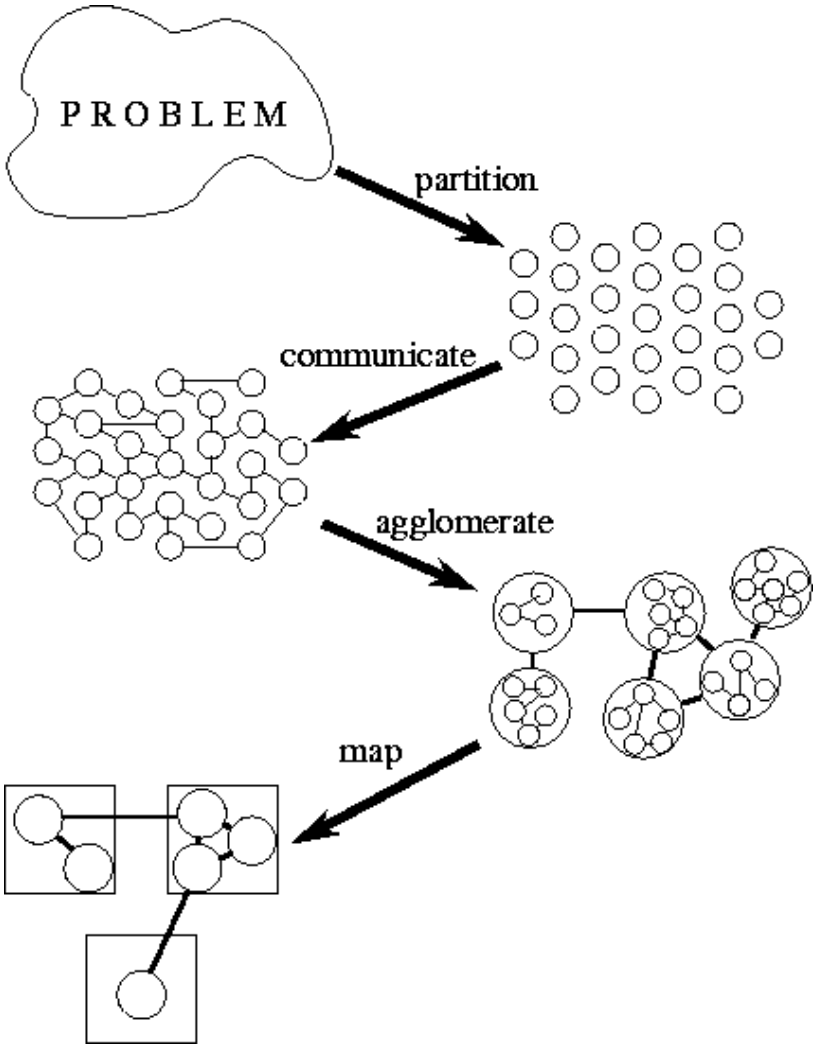
PCAM Methodology

Usually: Good parallel algorithm $\neq$ best sequential algorithm parallelized



PCAM Methodology

- *Partitioning*
  - The computation that is to be performed and the data operated on by this computation are decomposed into small tasks
  - Practical issues such as the number of processors in the target computer are ignored
  - Attention is focused on recognizing opportunities for parallel execution
- *Communication*
  - The communication required to coordinate task execution is determined
  - Appropriate communication structures and algorithms are defined
- Partitioning and Communication – concentration on concurrency, scalability

PCAM Methodology

- *Agglomeration*
  - The task and communication structures defined in the first two stages of a design are evaluated with respect to performance requirements and implementation costs
  - If necessary, tasks are combined into larger tasks to improve performance or to reduce development costs
- *Mapping*
  - Each task is assigned to a processor
  - Competing goals of maximizing processor utilization and minimizing communication costs
  - Mapping can be specified statically or determined at runtime by load-balancing algorithms
- Agglomeration and Mapping – concentration on locality and performance